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Direct Lithography for Regulating Multiple Properties of Organic Semiconductors via Photo-Crosslinkers

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Photo-crosslinkers, as materials with negative photoresist characteristics, are applied in the direct lithography process of organic semiconductor devices. Compared to commonly used organic lithography processes, the direct lithography process only requires three steps: spin-coating, exposure, and development, to achieve optical patterning of the organic semiconductor channel, eliminating the need for complex procedures such as protection, baking, etching, and transfer. Furthermore, during this process, the photo-crosslinker remains in the blended film, subsequently influencing the crystalline structure and morphology of the semiconductor film. These alterations will further impact multiple properties of the organic film. By utilizing this characteristic, the design of photo-crosslinker can effectively regulate and enhance the tensile properties, charge carrier mobility, stability, and dielectric properties of the device. This approach effectively enables high-precision patterning of organic integrated circuits while synchronously enhancing their various performance attributes. Although there are some relevant reports, a systematic summary and organization of the impact of these photo-crosslinkers on semiconductors remains absent. For the future development of direct lithography, systematic organization is crucial. Therefore, this review systematically classifies and summarizes the functions of photo-crosslinkers in organic thin film devices, providing guidance for the design of novel photo-crosslinkers in direct lithography and improve multiple properties of semiconductor devices.

1. Introduction

Compared with silicon electronics, organic electronics exhibit the advantages of adjustable conductivity levels, solution processability, intrinsic mechanical flexibility, and mixed ionic-electronic conductivity. These attributes make them well-suited for applications in soft actuators and robotics, flexible wearable devices, and implantable electronics.[1–5] The growing sophistication of organic semiconductors in chemical/electrochemical polymerization,[6,7] and solution processing[8] is driving their rapid progression toward large-scale, cost-effective applications.[9–11] Lithography, renowned for its high precision, scalability, and capability to produce complex patterns, effectively enhances the integration density and processing speed of chips, making it a core technique for the fabrication high-performance micro/nanoscale silicon-based integrated circuits.[12] Similarly, the performance of organic integrated circuits is also affected by the density of devices.[13,14] Therefore, utilizing precise patterning to reduce parasitic capacitance and crosstalk between adjacent devices will further promote the rapid development of organic electronics. Currently, direct capillary writing,[15] nano-imprinting,[16] inkjet printing,[17,18] and electrogelation patterning[19] have been utilized for patterning organic semiconductors. However, their inherent issues limit the further improvement of organic devices. Direct capillary writing is difficult to directly process viscous solution. Nano-imprinting has limited applicability regarding materials and device architectures (thermoplastic materials require high temperatures and pressures).[16] Additionally, Inkjet printing is a sequential process where ink is locally heated, generating rapidly expanding vapor bubbles that subsequently eject ink droplets.[20] It relies on the smooth flow and precise ejection of ink through the nozzle. The properties of ink, such as viscosity, surface tension, and particle size, directly affect print quality. Different ink requires distinct printing parameters.[21,22] Electrogelation patterning generates patterns by regulating the high ionic strength regions near the substrate surface, a process that typically requires the polymer to contain ions. In contrast, lithography has been a well-established and widely applicable technology in the silicon electronics industry, achieving high resolution and throughput.[23,24] However, the commonly used organic lithography process is not applicable to the patterning of most organic-based semiconductors. The complex processes involved, including deposition, baking, etching,

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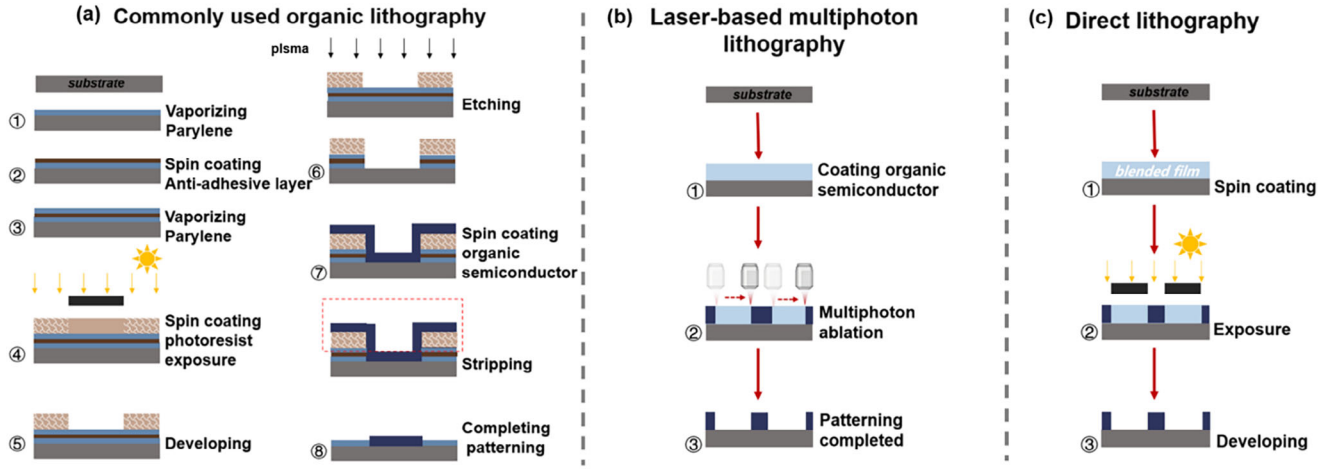


Figure 1. The patterning process of commonly used organic lithography a), laser-based mulitiphoton lithography b) and direct lithography c).

and stripping, can cause irreversible physical or chemical damage to the organic molecules, thus limiting the fabrication of high-performance organic integrated circuits.[25] (**Figure 1a**) Recently, laser-based multiphoton lithography has been applied to the micro/nanofabrication of organic semiconductors, enabling flexible control over patterning without the need for masks.[26–28] (Figure 1b) However, the laser exposure area may generate a significant heat-affected zone, which will cause damage to nearby polymer films. Additionally, selective layer-by-layer patterning poses a challenge in the fabrication of multilayer structures.

In contrast, direct lithography employing organic photo-crosslinkers, involving only three steps of spin-coating, exposure, and development (Figure 1c), is suitable for the precise patterning of organic semiconductor devices.[29,30] Photo-crosslinkers contain two or more photoactive groups and initiate a cross-linking reaction upon exposure to light of specific wavelengths. After cross-linking, the solubility of the organic semiconductor undergoes significant changes after cross-linking. This difference in solubility enables the patterning process of organic semiconductors to be carried out through solution processing as well. This process necessitates neither costly equipment nor stringent experimental conditions. Notably, the photo-crosslinker remains in the blended film after development, playing a crucial role in determining the overall performance of the device. The retention of photo-crosslinkers will affect various properties of the blended film, including tensile strength, stability, charge mobility, and dielectric constant. Therefore, by optimizing the molecular design of the photo-crosslinker and the direct lithography process, multiple properties of organic semiconductors can be enhanced, thus further promoting the advancement of organic electronics technology. The lack of reviews on this topic prompted us to embark on this paper, aiming to provide a comprehensive outlook on the development and application of photo-crosslinkers in organic semiconductor technology via direct lithography. The scope of this article covers the types of existing photo-crosslinkers, the mechanisms of cross-linking, and the details of condition for direct lithography. Based on the distinct effects of photo-crosslinkers in the blended films, we have conducted a classification and offered unique insights into design strategies. As a review paper, this article will further enhance researchers' understanding of direct lithography and provide valuable guidance for the design of novel photo-crosslinker and their application in organic semiconductors.

2. Structure of Photo-Crosslinkers and Cross-Linking Mechanisms

Understanding the molecular structure and corresponding cross-linking mechanisms of photo-crosslinkers is of great importance in exploring their specific effects in thin films during direct lithography. After summarizing the structures of existing cross-linker, we classify photo-crosslinkers into three components: the central core, the linking chain segment, and the photoactive group. The central core refers to the molecular backbone or atom located at the center of the photo-crosslinker, which determines the number of cross-linking sites and the spatial structure of the photo-crosslinker (1D, 2D, or 3D). The linking chain segment is the molecular chain that connects the central core to the photoactive groups, influencing the molecular size and solubility of the photo-crosslinker. The photoactive group, which undergoes chemical reactions upon exposure to light of a specific wavelength range, is a key factor in initiating the photo-crosslinking process.

This review classifies photoactive groups into five categories based on their reaction mechanisms. (**Figure 2**) The cross-linking of oxetane and glycidyl occurs via a cationic ring-opening mechanism (Figure 2a).[31,32] In the presence of a photoacid generator, ultraviolet light activates the generator to produce protons. These protons initially react with oxetane and glycidyl to form an oxonium intermediate. Subsequently, an electrophilic attack by another ethylene oxide molecule occurs at the α-carbon position of the oxonium intermediate, leading to the completion of the ring-opening and polymerization reactions.[33] Typically, elevated temperatures are required to advance the cationic ring-opening reaction, therefore a post-exposure annealing process is often incorporated in practical applications.[31,34]

Photoactive groups that generate carbon radicals upon light exposure belong to the categories of free radical polymerization and free radical coupling. Acrylates and diacetylene photo-crosslinker undergo cross-linking through free radical polymerization

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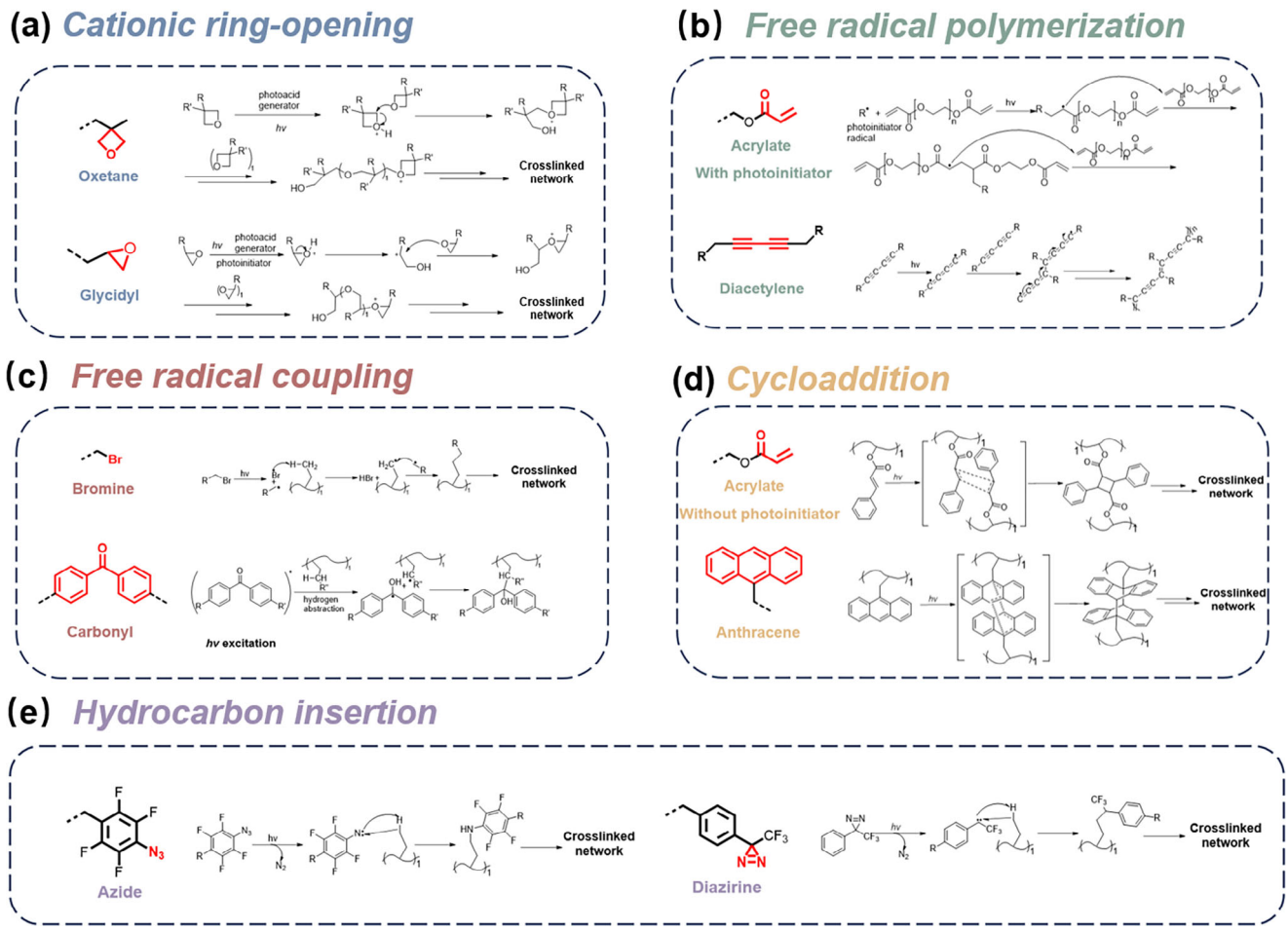


Figure 2. Structures and cross-linking mechanism of common photoactive groups.

(Figure 2b). This process usually requires a photoinitiator that generates free radicals upon illumination to initiate free radical polymerization. The entire process consists of three stages: chain initiation, chain propagation, and chain termination.[35–38] The cross-linking of bromine-based and benzophenone-based cross-linkers proceeds through a free radical coupling mechanism (Figure 2c).[39,40] Ultraviolet (UV) exposure at specific light wavelengths induces homolytic cleavage of the carbon-bromine covalent bond, generating bromine radicals that abstract protons from adjacent polymer side chains to form alkyl radicals. These alkyl radicals then couple.[41] At a wavelength of 365 nm, the carbonyl group of benzophenone can abstract a hydrogen atom, transforming into a ketyl radical. This radical then forms a covalent with an adjacent alkyl carbon, resulting in cross-linking. In the absence of a photoinitiator, acrylates (285 nm) and anthracene (365 nm) undergo cross-linking via a [2+2] cycloaddition reaction (Figure 2d).[42–44] α, β-unsaturated carbonyl compounds can be easily excited to a triplet state by direct irradiation or in the presence of a sensitizer. Upon photoexcitation, this olefin can react with an olefin in its ground state, forming a 1,4-biradical, which then undergoes ring closure to yield the cyclobutane product.[45] These types of photoactive groups typically undergo cross-linking with themselves and typically influence the solubility of semiconductors by encapsulating them. Due to their indirect cross-linking with the semiconductors, their efficiency tends to be low.[46] During the cross-linking process, photoactive groups that generate carbene or nitrene intermediates are classified as hydrocobalt insertion reactions.[40] Azide, diazo, diazirine groups cross-link through C-H insertion after generating nitrenes or carbene upon light exposure.[29,47,48] Cross-linkers that directly insert sp^3 C-H with semiconductors typically exhibit higher cross-linking efficiency (Figure 2e). Using azide as photoactive group, after photopatterning (254 nm) using a shadow mask inside a glove box, the polymer blended films were completed the cross-linking and patterned process.[49] When the diazirine is exposed to UV light (365 nm), the photoactive group in the cross-linker releases nitrogen and forms carbene. The carbene is then quickly inserted into the C−H bond of the polymer to achieve cross-linking[43] Diazo compounds can direct insertion into the C-H bonds of non-conjugated alkyl side chains in polymers under light exposure (365 nm) combined with thermal heating.[40,48,50] The lithographic performance of cross-linker can be evaluated using resolution, sensitivity, and line edge roughness.[51] Resolution, typically characterized by the smallest imageable line width, represents the minimum feature size that can be clearly defined on a substrate.[51] Line edge roughness, observed along the edges

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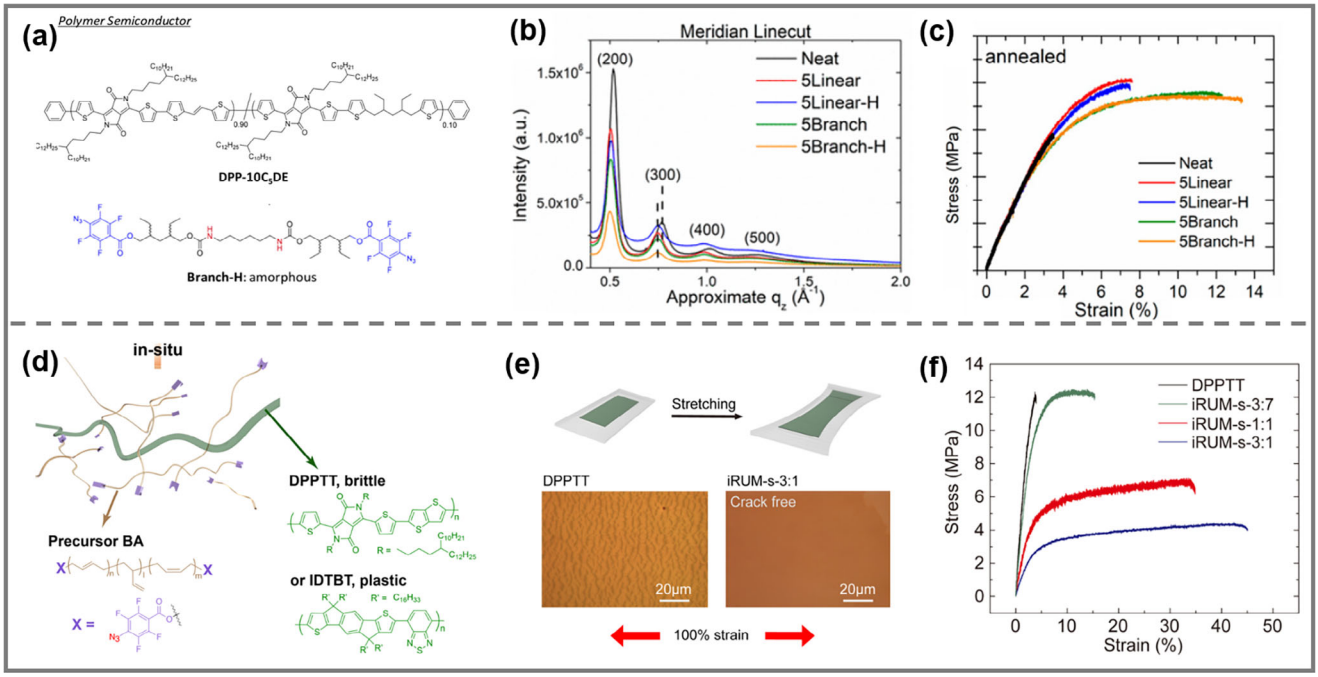


Figure 3. a) Chemical structure of the cross-linkers and the DPP-based polymer. b) Polarization-corrected meridian ($q_{xy} = 0$) line cuts of all cross-linked films. c) Stress–strain curve of annealed polymer-cross-linker blends. Reproduced with permission.[56] Copyright 2019, American Chemical Society. d) Chemical structure of the cross-linkers and the polymer. e) Schematic illustration of the stretching of the semiconductor film on a supported PDMS substrate. Optical microscope images of a neat DPPTT film and an iRUM-s-3:1 film under 100% strain. f) The representative stress-strain (engineering) curves for DPPTT and iRUM-s-x:y films obtained from pseudo-free-standing tensile tests. Reproduced with permission.[57] Copyright 2021, Springer Nature.

of lithographically created lines or patterns, indicates the process's capability to produce fine details.[52] Sensitivity, defined as the energy required for a photo-crosslinker to achieve a specific patterning level, reflects its responsiveness to light exposure.[51]

The efficiency of cross-linking, and thus the lithographic performance, depends on several factors, including the structure of the polymer and cross-linker, the concentration of the cross-linker, the exposure wavelength, and the exposure dose.[50] Therefore, in practical applications, it is crucial to optimize these conditions for each specific cross-linker and polymer in order to determine the most effective patterning parameters. When a photoacid generator or photoinitiator is used in the cross-linking, the exposure wavelength will be determined by the photoinitiator, not the cross-linker. Moreover, the electronic properties of groups near the photoactive groups may influence the exposure wavelength.[40] The appropriate exposure wavelength can be determined by analyzing changes in the UV-visible absorption peaks of cross-linker before and after exposure. The exposure dose and cross-linker concentration can be optimized based on the retention rate or resolution of the thin film after exposure and development. Additionally, the final device performance after development can further inform the optimal concentration of the cross-linker. Regardless of the cross-linker type, cross-linker left in the film would alter its physical or chemical properties, affecting device performance. Understanding these properties is crucial for designing and applying cross-linkers more rationally, as well as for optimizing the performance of semiconductor films.

3. Tensile Properties

Low-modulus and stretchable organic electronic materials hold significant promise for next-generation wearable and implantable applications. It is generally considered that achieving both excellent charge transport and stretchability in pristine films is mutually exclusive. Long-range ordered crystalline domains are generally required for high charge carrier mobility, whereas a substantial amount of disordered domains is necessary for flexibility and stretchability.[53] Consequently, most high carrier mobility polymer semiconductors tend to fracture under low strain.[54] The introduction of photo-crosslinker can effectively improve this situation. After blending cross-linkers for cross-linking, the polymer chains between cross-linking sites undergo conformational changes during stretching. Upon strain release, the matrix returns to its original state,[53,55] which contributes to improved stretchability of the film.

According to previous reports, the use of azide cross-linkers with tertiary carbon side chains enhances the stretchability of DPP-10C5DE (**Figure 3**a–c).[56] Line-cut Grazing Incidence Wide Angle X-ray Scattering (GIWAXS) patterns revealed that, compared to pure films and films cross-linked with linear cross-linkers, the semiconductor with a branched cross-linker has the largerest full width at half maximum (FWHM) and the largest lattice spacing, indicating the lowest crystallinity (Figure 3b). This reduction in crystallinity and aggregation level contributed to better tensile performance. After annealing, the fracture strain increased from 2.5 ± 1.0% for the pure film to 8.0 ± 2.0% for

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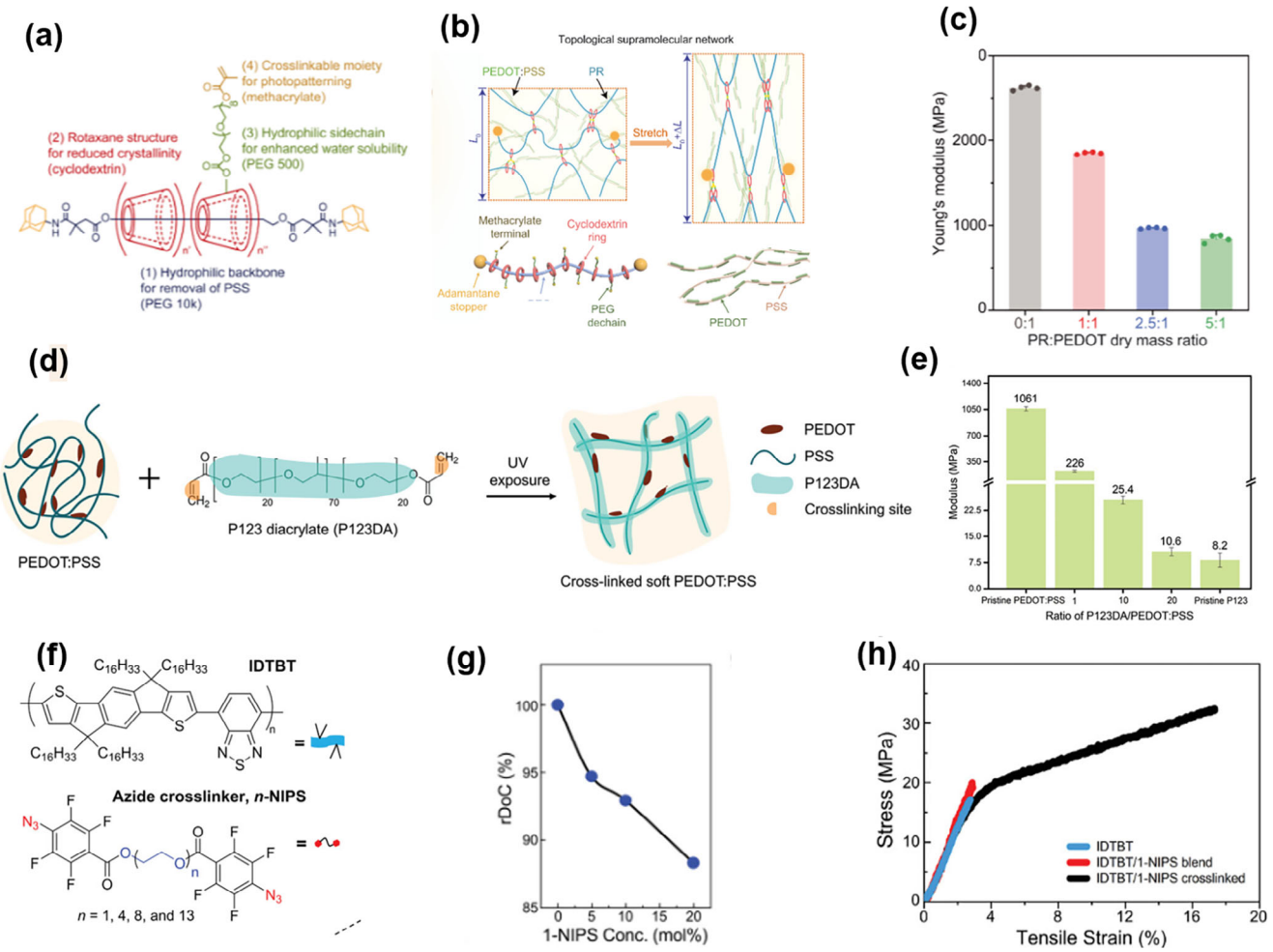


Figure 4. a) Chemical structure of the PR-based cross-linkers. b) Schematic diagram illustrating the intrinsically stretchable topological supramolecular network with the key molecular building blocks of PR monomers. c) Statistical comparison of Young's moduli measured by nanoindentation for different TopoE films indicating that PR can reduce the overall film stiffness. Reproduced with permission.[60] Copyright 2022, The American Association for the Advancement of Science. d) Schematic illustration of P123DA/PEDOT:PSS cross-linked network. e) Average DMT moduli of P123DA/PEDOT:PSS. Reproduced with permission.[61] Copyright 2024, Wiley-VCH. f) Chemical structures of IDTBT and *n*-NIPS. g) Estimated relative degree of crystallinity (rDoC) of the cross-linked IDTBT films according to 1-NIPS concentration. h) Stress–strain curves of IDTBT films upon mechanical strain, measured by the film-on-water testing method. Reproduced with permission.[62] Copyright 2023, Wiley-VCH.

the cross-linked film, showing a 320% improvement, while the crack-onset increased from 25% to over 100% (Figure 3c). Similarly, the use of polybutadiene hydrogenated azide as a cross-linker can further improve the tensile properties of DPPTT.[57] (Figure 3d–f) The cross-linked elastic rubber matrix absorbs strain energy through conformational changes, while the covalent cross-linking sites provide restoring force, enabling the film to revert to its original state. The cross-linked film in-situ rubber matrix (iRUM-s-3:1 can stretch up to 100% strain without any visible cracks. In contrast, the neat DPPTT film shows dense cracks at 100% strain. (Figure 3e) According to the stress-strain curves, the fracture strain of the iRUM-s-3:1 film increased by nearly one order of magnitude compared to the neat film, rising from 5% to 45%, while its elastic modulus decreased by four-time (Figure 3f). In addition to polymer semiconductors, the iRUM method can also be applied to high-performance dielectric materials such as Styrene Ethylene Butylene Styrene. With the assistance of photo-crosslinkers, fully patterned elastic transistors can be fabricated on a single substrate, facilitating the integration of a monolithic flexible electronic skin.[58]

Poly (3,4-ethylenedioxythiophene)-poly (styrenesulfonate) (PEDOT:PSS) is one of the most successful conducting polymers due to its impressive chemical stability, excellent biocompatibility, and good electrical conductivity. However, the semi-crystalline nature of PEDOT:PSS results in poor intrinsic stretchability.[59] To satisfy the requirements of wearable electronic applications, adding cross-linkers represents one of the effective strategies to achieve highly stretchable PEDOT:PSS. Employing polyrotaxane (PR) as a backbone with cyclodextrin (CD) sliding rings as supramolecular photo-crosslinkers can further improve the stretchability of the film.[60] (**Figure** 4a) The sliding CD units inhibit crystallization while providing significant conformational freedom through its mechanical interlocking structure. (Figure 4b) As the PR content increases, the films exhibit

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improved stretchability and a lower Young's modulus, decreasing from >2500 MPa for the neat film to <1000 MPa for the cross-linked film, a reduction of nearly 60%. (Figure 4c) The use of this supramolecular cross-linker enables direct patterning of high-density, stretchable arrays, allowing for high-precision bioelectronic applications in the fragile rat brainstem. Another example of using cross-linker to enhance the stretchability of PEDOT:PSS is the direct cross-linking of PE123DA. (Figure 4d) Through Derjagun-Muller-Toporov (DMT) modulus measurements, it was found that as the ratio of P123DA increased, the modulus of the PEDOT:PSS electrodes decreased from 226 to 10.6 MPa. (Figure 4e) These results demonstrate that by adjusting the ratio of P123DA to PEDOT:PSS, the morphological structure and mechanical properties of the electrodes can be effectively controlled.[61] Using polar chain cross-linker is also an effective method to improve the tensile properties of organic semiconductor thin films. The polar chain cross-linker n-NIPS significantly improved the tensile properties of the organic semiconductor IDTBT (Figure 4f) after cross-linking.[62] The incorporation of 1-NIPS increased the content of disordered, amorphous regions in the semiconductor, which implise that the inclusion of 1-NIPS in IDTBT films leads to films that are easily deformed (Figure 4g). The IDTBT and IDTBT/1-NIPS blended films tear at ≈3% strain, whereas the IDTBT/1-NIPS cross-linked film can withstand up to 18% strain, resulting in a 6-fold increase in fracture strain (Figure 4h).

In practical applications, materials used in biosensors must possess sufficient flexibility to cover dynamically moving surfaces.[59] The examples mentioned above highlight the significant impact of photo-crosslinkers on the mechanical properties of conjugated polymer films. The photo-crosslinking strategy has brought new development opportunities to materials with poor initial tensile properties, effectively reducing the development costs.[53] By utilizing photo-crosslinkers, it is possible to achieve high-precision and complex logic circuits while maintaining the flexibility of the device, making it more suitable for emerging applications such as contact with biological tissues[13] and emerging flexible robotics.[63] However, the morphological changes resulting from the blending of cross-linkers may adversely affect the electrical performance of semiconductors.[64] To enhance the stretchability of films, a key focus should be placed on the design of the linking chain segment within the cross-linker. One effective strategy is to introduce flexible segments that allow for greater conformational freedom. For example, replacing rigid CCC (-C-) bonds with more flexible COC (-O-) bonds can facilitate the dissipation of strain energy, which may improve the overall stretchability of the films. Additionally, incorporating flexible spacers (such as hydrogen bond[65,66] or conjugated rigid molten rings[67]), including amide or amino groups, into the linking chain segment can facilitate the formation of internal entanglement structures within the material. This modification also strengthens the intermolecular interactions of the cross-linker, enhancing the stability and stretchability of the cross-linked films. Moreover, the introduction of larger side chains or asymmetric derivatives can further reduce crystallization tendencies and minimize the aggregation of rigid backbones in semiconductor. This, in turn, leads to fewer interfacial regions, thus improving the ductility of the blend films.[68]

Additionally, high-molecular-weight polymers tend to exhibit superior stretchability.[69,70] Therefore, designing polymer-based photo-crosslinkers may potentially enhance the stretchability of the cross-linked network. The incorporation of thermoplastic elastomer structures, such as Thermoplastic Polyamide Elastomer and Thermoplastic Vulcanizate or the integration of natural polymers (e.g., carbohydrate derivatives) into cross-linker holds promise for developing stretchable polymer based cross-linker suitable for direct lithography. While there have been studies on polymer cross-linkers,[51] the mechanical properties of these cross-linked networks have not been extensively explored. Further investigation into this area could provide valuable insights for the development of more stretchable and durable materials.

4. Device Stability

The stability of organic electronic devices is a major challenge faced in the industrialization process of organic semiconductors. The key factors affecting device stability include material diffusion during operation, charge carrier quenching by water and oxygen, and structural degradation induced by light and heat. In bulk heterojunction (BHJ) solar cells, blending photo-crosslinkers into the active layer of solar cells prevents fullerene diffusion and the formation of large microcrystals,[71] (**Figure 5a**) thus stabilizing the BHJ morphology After cross-linking the bulk heterojunction (SiIDT-BT/PC[70]BM) with the small molecule cross-linker DAZH (Figure 5b), optical microscopy reveals that the addition of DAZH significantly suppresses the crystallization of fullerene, compared to neat films[72] (Figure 5c). Atomic force microscopy further shows that the microstructure of the bulk heterojunction exhibits more fibrous characteristics, which is expected to enhance charge separation and transport. When subjected to 130 h at 85 °C under an inert atmosphere, the devices containing DAZH demonstrate improved stability, with a PCE retention rate exceeding 80%, compared to less than 60% for the pure films (Figure 5d). In addition to the improvement in stability, the Organic Photovoltage (OPV) devices using DAZH also show a synergistic enhancement in power conversion efficiency (PCE). For 10% wt DAZH, the PCE increases from 6.0% to 7.0% (relative to the polymer), indicating that the use of small molecule photo-crosslinkers is one of the effective strategies for achieving high-performance OPVs.

In addition to stabilizing the morphology, the cross-linking strategy also reduces the diffusion of water and oxygen within the film, further improving its stability. The four-armed photo-crosslinker 4BX, when used to cross-link both p-type and n-type materials, (**Figure 6a**) demonstrates enhanced stability in organic field-effect transistors (OTFT).[49] This improvement may be attributed to the slower diffusion of oxygen or moisture within the photo-crosslinked polymer films, which reduces the likelihood of carrier recombination inside the film. After 35 days, the hole mobility of P(DPP2DT-TVT) and the electron mobility of P(NDI3OT-Se2) are approximately one order of magnitude higher in the cross-linked films than in the neat films (Figure 6b,c). Furthermore, some semiconductor films become denser after cross-linking, exhibiting improved hydrophobic properties, which may contribute to enhanced device stability. Furthermore, this 4BX can be used to pattern dielectrics (Poly(methyl methacrylate)

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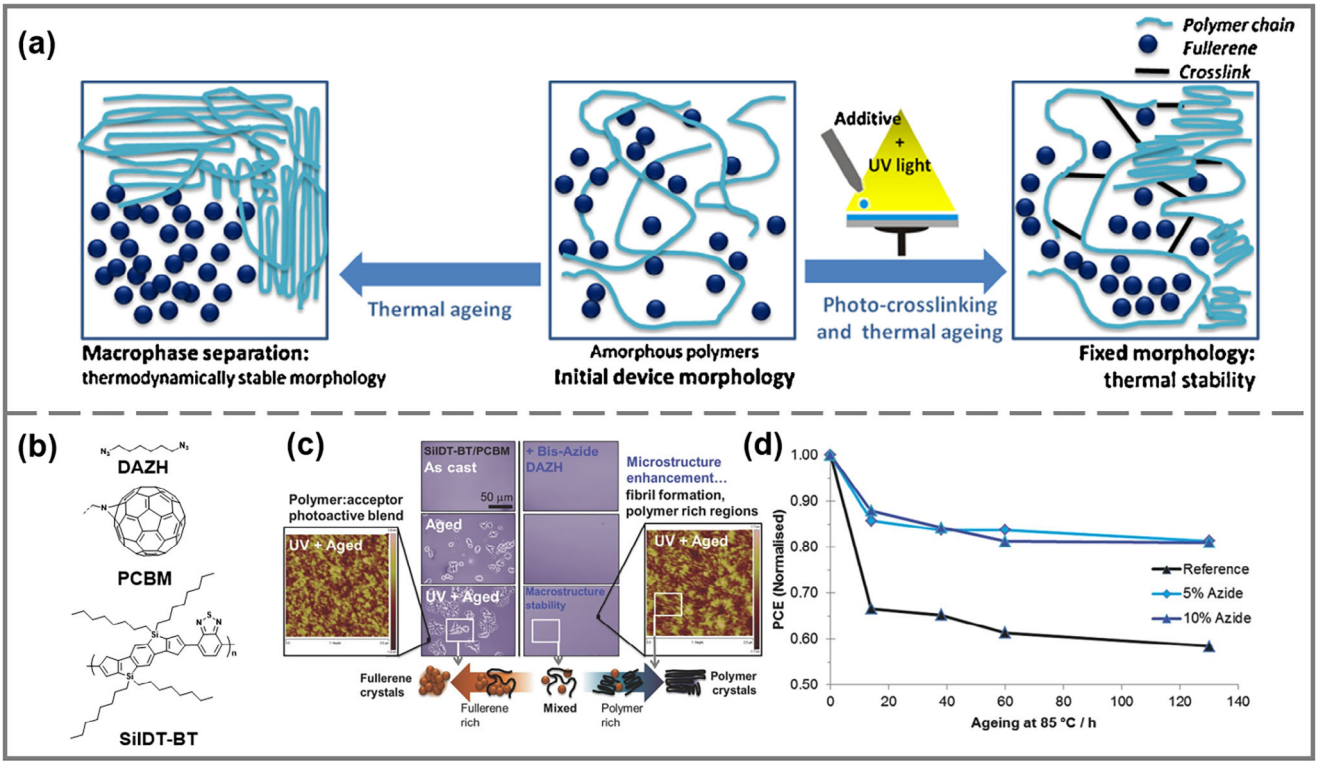


Figure 5. a) Additive cross-linking conveys thermal stability to optimized blends inhibiting macrophase crystallization (shown here for a portion of the substrate) Reproduced with permission.[71] Copyright 2015, Elsevier. b) Chemical structure of the cross-linkers, donor, and acceptor in OPV. c) Small molecule additives confer macrostructure thermal stability, through suppressed fullerene crystallization (central optical microscopy images), with the bis-azide DAZH additionally enhancing the microstructure through fibril formation (flanking AFM images). d) Photo-crosslinking SiIDT-BT/PC[70]BM organic solar cells with bis-azide DAZH, with 30 min UV curing time. Reproduced with permission.[72] Copyright 2015, Wiley-VCH.

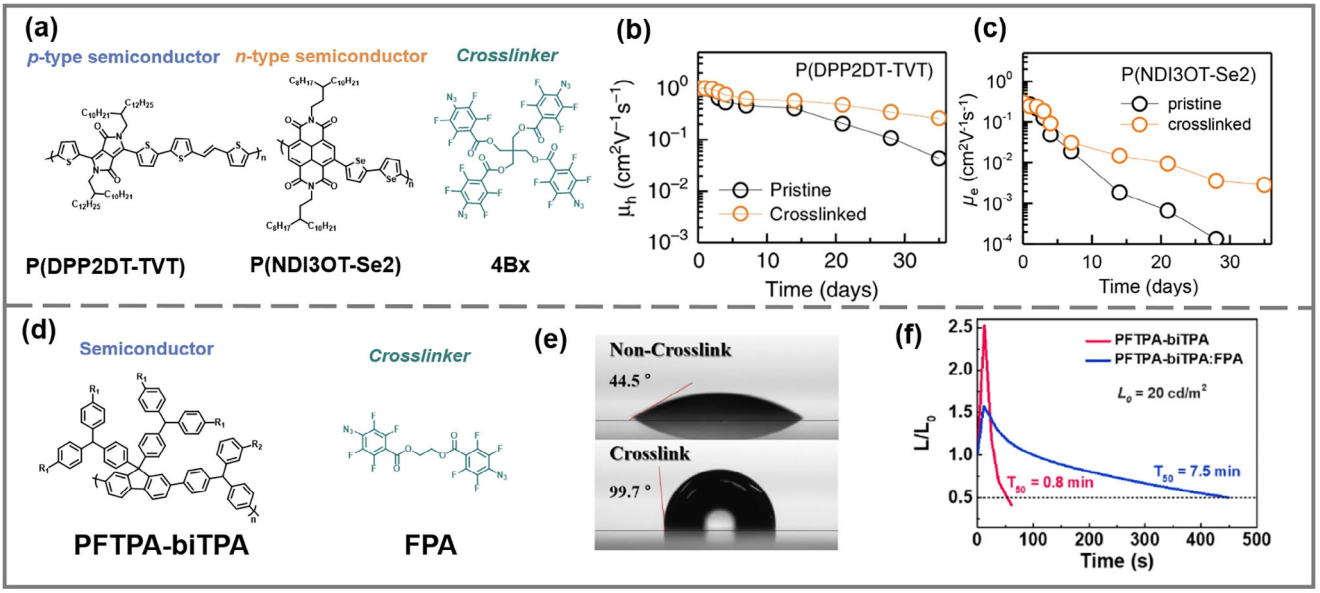


Figure 6. a) Chemical structure of the P(DPP2DT-TVT), P(NDI3IOT-Se2), and 4Bx. b) Comparison of the hole and electronic mobilities over time for pristine and cross-linked P(DPP2DT-TVT) (b) and P(NDI3IOT-Se2) c), respectively. Reproduced with permission.[49] Copyright 2020, Springer Nature. d) Chemical structure of the PFTPA-biTPA and FPA.[74] e) Contact angles of the non-cross-linked and cross-linked PFTPA-biTPA. f) Operational stability of the neat film and cross-linked film-based PeOLED. Reproduced with permission.[74] Copyright 2023, Springer Nature.

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and electrodes (Ag NP), enabling full photolithographic patterning for transistors and complementary logic circuits. This approach is expected to pave the way for the fabrication of more complex miniature circuits.[73] The cross-linked films were utilized as hole transport layers (HTLs) in OLEDs. (Figure 6d) The cross-linked PFTPA-biTPA films exhibited significantly higher solvent stability, better external quantum efficiency (EQE), and improved operational stability compared to the neat films.[74] The contact angle of the cross-linked films with water was notably higher than that of the neat films (increased from 44.5° to 99.7°), (Figure 6e) indicating that the cross-linking process enhanced the hydrophobicity of the films. This could protect the adjacent perovskite layer from moisture degradation. Atomic force microscopy (AFM) imaging revealed that the cross-linked films were denser and more uniform, which contributed to improved hole transport and reduced leakage current. The enhanced solvent resistance of the cross-linked films also improved the interface quality and reduced defects, resulting in devices with cross-linked HTLs exhibiting over 9 times greater operational stability than those with neat HTLs. (Figure 6f) This suggests that cross-linking may be a critical strategy for addressing stability issues in solution-processed multilayer devices. Although organic electronic materials show promise for lightweight and flexible devices, they generally exhibit lower stability than inorganic materials.[75] To enhance stability, covalent bonds can be formed between the photo-crosslinkers and the organic semiconductors. Compared to vapor deposition and thermal cross-linking techniques,[76–78] photo-crosslinking offers milder conditions, minimizing thermal degradation of underlying layers.[79,80] Moreover, photo-crosslinkers can be employed to create dynamic cross-linking networks, endowing semiconductors with self-healing capabilities.[81] Although organic materials generally exhibit lower stability than inorganic materials, their stability can be improved through the formation of covalent bonds. The selection of photoactive groups that can directly cross-link with the semiconductor is crucial for enhancing the stability of the semiconductor cross-linking network. Additionally, the choice of linker segments in the cross-linker is also vital. Specifically, selecting linker chain segments that are compatible with the semiconductor can improve the uniformity and efficiency of cross-linking within the semiconductor film. For instance, incorporating longer alkyl chains or increasing branching in the linker can enhance the miscibility of the cross-linker with the semiconductor. This improves phase compatibility, helping to reduce phase separation and molecular aggregation, which, in turn, enhances the overall stability of the device.[82]

Additionally, incorporating highly reducing species in the linker chain segments, such as 1,2-diol or catechol, can be beneficial. These species can remove reactive oxygen species within the cross-linked network through oxidative sacrifice. This process aids in protecting the n-type polymer molecules from oxidative damage and can passivate potential electronic traps, thus enhancing charge transport stability.[83] Meanwhile, using of cross-linkers and direct lithography allows the thin films to achieve good solvent resistance, which facilitates the fabrication of multilayer heterojunction devices. Consequently, a patternable passivation layer can also be introduced during device fabrication to protect the active layer from exposure to water and active oxygen.

In fact, there is limited research on the changes in the cross-linked network during device operation or under varying environmental conditions. This knowledge gapnecessitates further investigation, as it may pave the way for the design of more stable organic semiconductor layers for device.

5. Carrier Mobility and Conductivity

Electronic and hole transport properties are crucial parameters for evaluating the performance of organic electronic devices. Carrier mobility is influenced by various factors, including crystal packing distance, crystal size, orientation, order, and surface morphology of the thin film. The introduction of cross-linker allows for the regulation of these factors. One of the best examples of cross-linkers influencing mobility is the use of ethylene glycol-based cross-linkers in PEDOT:PSS. The strong interactions between the PEDOT and PEG linking segment lead to a transformation of the PEDOT:PSS structure from a core-shell structure to more extended chains (**Figure 7a**).[84] As the amount of Polyethylene glycol (PEG) cross-linker increases, intermolecular interactions intensify, reducing the π–π stacking distance of PEDOT (Figure 7b). This more compact stacking and the formation of a more interconnected nanofiber network significantly enhance the conductivity of PEDOT:PSS. The conductivity of the original PEDOT:PSS can be increased from 12.0 ± 3.1 S m^{-1} to 3.15 ± 0.68 kS m^{-1} for PEDOT:PSS with PEG cross-linker(Figure 7c). The utilization of PEG photo-crosslinker and diazirine cross-linker enables the fabrication of fully polymeric elastic transistor arrays with a device density of ≈42,000 transistors per cm^2. This high-density fabrication technique also enables the creation of various complex electronic devices for practical applications, including biosensing[85–87] and organic light-emitting.[88] This strategy is not only applicable to improving the packing and mobility of polymers, but similar effects have also been observed in certain small molecule semiconductors. X-ray diffraction (XRD) results show that cross-linker can serve as effective bridging materials for the fullerene-based LT-S9079 film (Figure 7d).[89] Cross-linking of LT-S9079 results in a more ordered structure. The 2D GIWAXS patterns show more ordered lattice stacking and stronger π–π stacking (Figure 7e,f). The formation of a rigid, interconnected network reduces electron-photon interactions and recombination energy, which enhances carrier transport. As a result, the conductivity of LT-S9079 increases significantly from 3.2×10^{-9} S cm^{-1} (neat film) to 4.5×10^{-5} S cm^{-1} (cross-linked film) (Figure 7g).

The use of 3D multi-arm cross-linkers has a more significant effect on improving the packing of organic polymers. The SP-1, composed of a tri-armed acrylate cross-linker and PTDPPTFT, (**Figure 8a**) forms a unique nanoscale interpenetrating structure. The aggregation structure of SP-1 was confirmed using AFM and Transmission Electron Microscopy (TEM).[90] SP-1 exhibits nanoscale phase separation and forms a uniform interpenetrating structure with bundled aggregates, whereas pure PTDPPTFT4 and PTDPPTFT4-MA, with photoactive groups on the side chains, form uniform thin films. The schematic diagram is shown in theFigure 8b,c. GIWAXS results show that the cross-linked SP-1 possesses a more compact and higher-order π–π stacking, as illustrated in the schematic. This structural enhancement leads to a higher mobility for SP-1 (> 0.9 cm^2 V^{-1} s^{-1}), while

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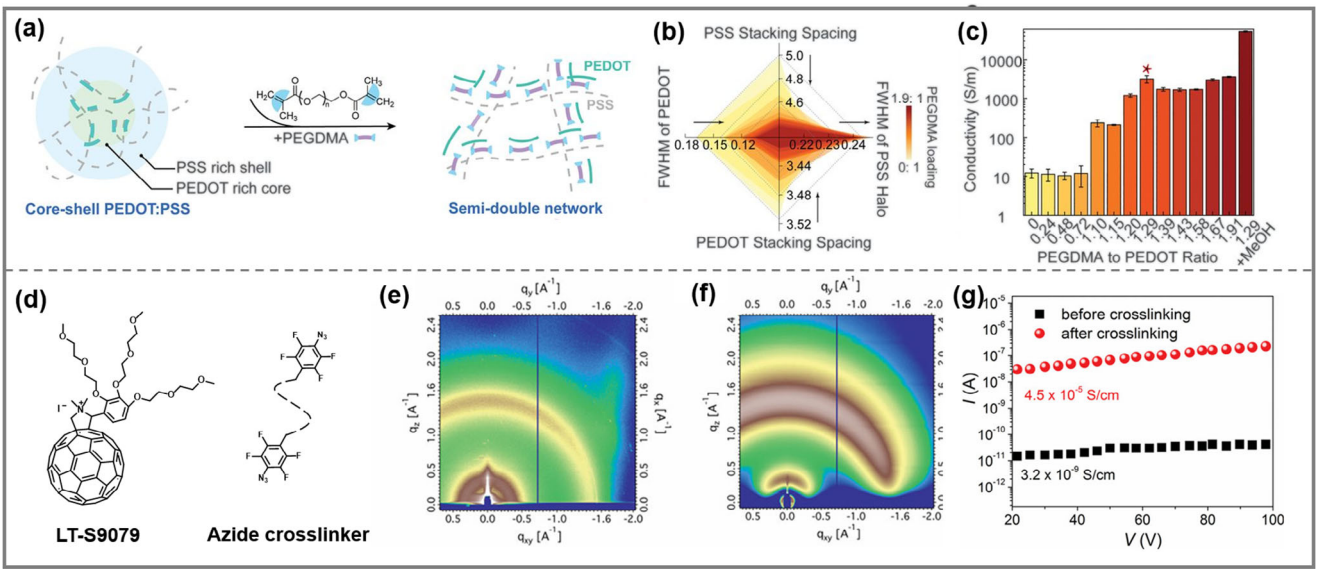


Figure 7. a) Mechanism of direct optical lithography of PEDOT:PSS by selective cross-linking of PEGDMA to form a double network that is insoluble in water. b) Radar chart displaying PSS π–π stacking distance, PEDOT π–π stacking distance, full width at half maximum (FWHM) of PSS scattering halo, and FWHM of PEDOT scattering halo determined from GIXD of UV-exposed PEDOT:PSS/PEGDMA films. c) Dependence of UV-exposed PEDOT:PSS/PEGDMA film conductivity on proportion of PEGDMA. Reproduced with permission.[84] Copyright 2021, The American Association for the Advancement of Science. d) The chemical structure of LT-S9079. XRD of e) before and f) after cross-linking of LT-S9079 films. g) *I*–*V* curves of before and after cross-linking of LT-S9079 films. Reproduced with permission.[89] Copyright 2017, Taylor & Francis.

PTDPPTFT4-MA exhibits a lower mobility (0.383 $cm^2\ V^{-1}\ s^{-1}$) (Figure 8d). The improvement in stacking allows SP-1 to achieve the highest mobility for a photo-crosslinkable organic semiconductor. In addition to improving the electrical performance of the devices, SP-1 also enables the realization of fully photolithographic organic circuits, facilitating the fabrication of organic inverters and high-density transistor arrays with a density of up to 1.1×10^5 devices cm^{-2}. By combining conjugated polymer films with thiol-ene chemistry, the mobility of semiconductor films can be significantly enhanced after the thiol cross-linker reacts with the elastomer under UV irradiation to form a cross-linked network.[91] (Figure 8e) This cross-linked network exhibits a

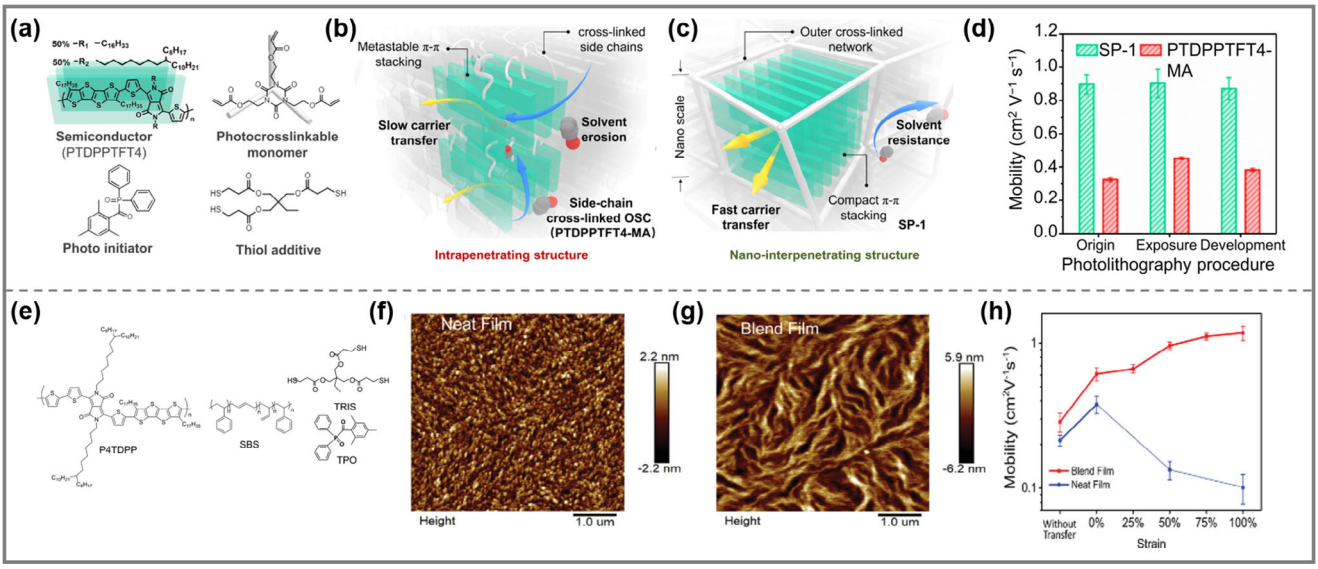


Figure 8. a) Schematic diagram showing the photo-crosslinking of SP-1. Schematic illustrations of b) nano-interpenetrating and c) intrapenetrating structures. d) Mobilities of SP-1 and PTDPPTFT4-MA during lithography procedures. Reproduced with permission.[90] Copyright 2021, The American Association for the Advancement of Science. e) Chemical structure of conjugated polymer P4TDPP, poly(styrene-butadiene-styrene) (SBS), and trimethylolpropane tris(3-mercapto propionate) (TRIS). AFM tapping height images of f) P4TDPP neat film and g) P4TDPP/SBS blend film under UV exposure for 1 min. h) The mobility of the neat film and the blend film at different strain ratios. Reproduced with permission.[91] Copyright 2023, Wiley-VCH.

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distinct nanofiber morphology. AFM images reveal that the neat P4TDPP film has a smoother surface with smaller aggregates, whereas the polymer blend film displays a fibrous morphology with larger crystalline grains (Figure 8f,g). The more ordered arrangement of the polymer chains in the blend facilitates better charge transport, resulting in improved charge carrier mobility in both the initial and stretched states (Figure 8h). In addition, there are more examples of using cross-linkers to enhance device mobility. SU8 can promote the aggregation of DPP during thin film deposition, forming fiber-like π–π stacking and polymer chain accumulation, which is beneficial to enhance carrier mobility.[92] The 3D acrylate cross-linker can form a uniform nanoparticle interpenetrating network embedded with perovskite quantum dot (PQD) during the cross-linking process. This nanostructured interpenetrating network ensures lithographic performance and high mobility.[93] Furthermore, PQD nanobatteries can be embedded into lithographic organic semiconductors (OSCs) to fabricate organic optoelectronic transistors. Using lithography, a full-frame ultra-large-scale integration chip with 27 million interconnected pixels has been successfully produced, reaching the level of the latest commercial camera chips.[94] Utilizing azide cross-linkers, chemically robust dielectric polystyrene films can be prepared. The chemical stability of these cross-linked films creates a favorable environment for the growth of semiconductor crystals, resulting in highly crystalline solution-processed semiconductor films on polystyrene dielectric layers. This improvement enhances the field-effect mobility of OFETs.[95] In general, cross-linkers enhance mobility by controlling the morphology and phase separation, creating ordered structures, and regulating crystal packing. Therefore, the design of cross-linkers should focus on both the central core.

The improvement of semiconductor mobility and the high integration of devices will reduce energy consumption and enhance signal processing and response speed in practical applications. This holds great potential for the development of advanced applications, including neuromorphic devices[96,97] and flexible electronic skin.[98,99] The central core of the cross-linker significantly influences the geometry and overall structure of the network. For example, using 2D covalent organic frameworks (COFs) or 3D cage-like structures as the central core may enhance the crystallinity and packing of the cross-linker. This could contribute to more orderly packing of the semiconductor, thus improving charge transport. In conjugated polymers, the use of asymmetric conjugated building blocks is a valuable strategy to tune both the energy levels and the packing of the polymer.[100,101] Asymmetric cores in cross-linkers could similarly influence the charge transport characteristics by modifying the alignment and packing of the semiconductor crystals in the cross-linked network. Despite these advancements, charge transport in semiconductors remains a complex process that is highly sensitive to changes in both the semiconductor structure and the cross-linker design. Even minor modifications to the cross-linker or the semiconductor can significantly impact the material's overall performance. Therefore, ongoing research into more versatile and finely tuned cross-linkers is essential for optimizing charge transport and achieving better semiconductor performance in various applications.

6. Dielectric Properties

In OFETs, achieving low power consumption and stability, the active layer requires high stability and charge mobility, while the dielectric layer requires a high dielectric constant and excellent insulating properties. The function of dielectric layer is similarly to a capacitor, requiring a high dielectric constant (κ), excellent dielectric strength, high mechanical strength, and optimal surface properties. Photo-crosslinkers, when blended and cross-linked with insulating polymer, produce denser and defect-free films. This prevents charge flow between the gate and the organic semiconductor, reduces the loss factor, and enhances the film's insulating properties. Cross-linking of cyclic olefin copolymer (COC) with azide-based photoreactive groups shows a reduction in dielectric leakage current.[102] (**Figure 9a**) As the number of photoreactive groups in the cross-linker increases, the leakage current decreases further. The leakage current density of the pure COC is 8.10×10^{-7} A cm^{-2} at 2 MV cm^{-1}. In contrast, the devices with 2F, 3F, and 4F COC cross-linked films show much lower leakage currents, with values of 3.94×10^{-7}, 9.16×10^{-8}, and 8.22×10^{-10} A cm^{-2}, respectively (Figure 9b). Additionally, testing on P-type devices (C10-DNTT) revealed even lower leakage currents (Figure 9c). The improved dielectric insulating properties may be attributed to the increased number of reactive sites in the FPA cross-linker, which leads to stronger interactions with the C—H bonds in the COC material, resulting in tighter molecular bonding and enhanced intermolecular forces. This causes the film to become denser and more insulating. The stable dielectric layer formed after cross-linking can also be successfully applied to the printing of solution-processed semiconductor carbon nanotubes and flexible organic field-effect transistors.[103,104] Through UV induced cross-linking, amorphous regions can be created near the crystalline domains, thus enhancing the dielectric performance of polymer materials through interfacial polarization. Specifically, PVDF-TrFE was used as the matrix, with BTNPs serving as the cross-linker, to prepare the cross-linked nanocomposite MPMS@BT/PVDF-TrFE under UV irradiation.[105] (Figure 9d) As shown in Figure 9e, the dielectric constant increases progressively with the increasing content of MPMS@BTNPs, rising from 10.62 to 15.57. This is attributed to the introduction of MPMS@BT NPs, which creates interfaces between the MPMS@BTNPs and the matrix. These interfaces lead to enhanced interfacial polarization, resulting in a moderate increase in the dielectric constant. Additionally, it was observed that as the irradiation dose increases, the crystallinity of the MPMS@BT/P(VDF-TrFE) nanocomposite decreases from 32.1% to 19.0%. (Figure 9f) This decrease is due to the formation of the cross-linked network during irradiation. Although crystallinity negatively correlates with the irradiation dose, the dielectric constant reaches its peak at an irradiation dose of 100 kGy. (Figure 9g) The increase in dielectric constant is primarily due to the interfacial polarization occurring at the additional interfaces between the crystalline and amorphous regions induced by cross-linking. However, the decrease in dielectric constant can be attributed to the reduced crystallinity and the structural damage to the molecular, which leads to the formation of more defects. Moreover, cross-linking could enhances adhesion to the substrate by forming covalent bonds, which increases energy density, and

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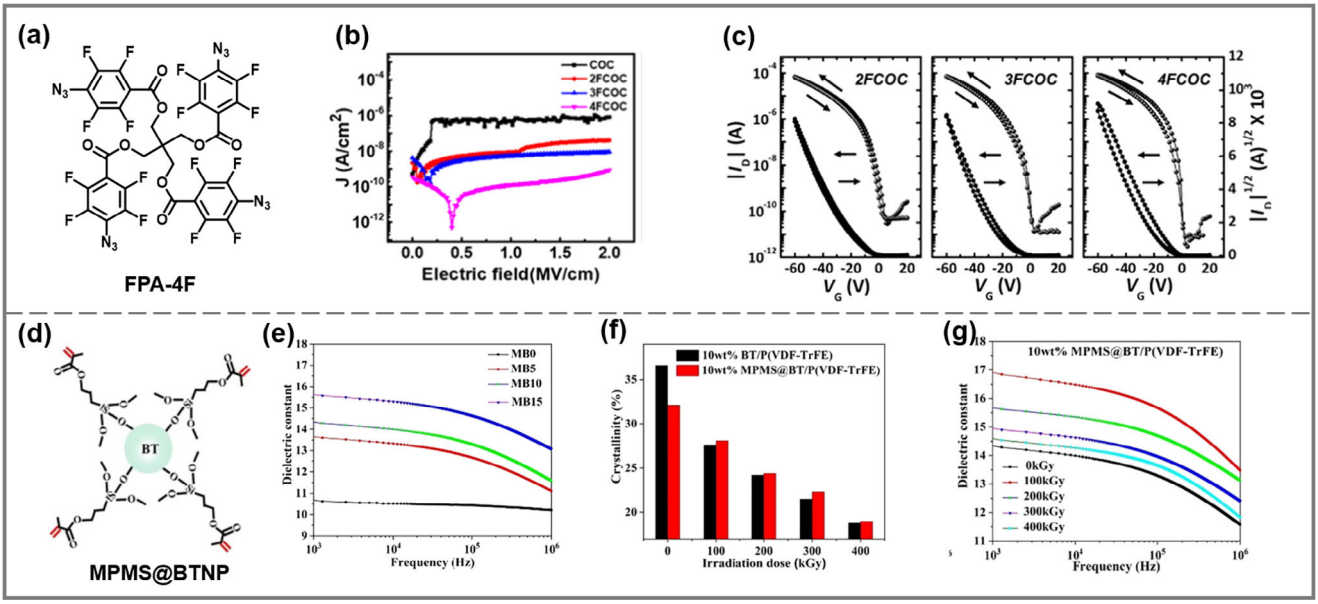


Figure 9. a) Chemical structure of the FPA-4F Series. b) *J* of pristine and cross-linked dielectric COC films. c) Electrical characteristics with transfer curves and output curves for C_{10}-DNTT based devices prepared using cross-linked dielectric layers. Reproduced with permission.[102] Copyright 2021, American Chemical Society. d) Chemical structure of the MPMS@BTNPs. e) Variation of relative permittivity with the frequency from 10^3 to 10^6 Hz of P(VDF-TrFE) with the content ranging from 0 to 15 wt.%. f) Crystallinity calculated by DSC thermograms of 10 wt% BT/P(VDF-TrFE) films and 10 wt.% MPMS@BT/P(VDF-TrFE) films with irradiation doses from 0 to 400 kGy. g) 10 wt.% MPMS@BT/P(VDF-TrFE) films with frequencies ranging from 10^3 to 10^6 Hz. Reproduced with permission.[105] Copyright 2021, Elsevier.

the cross-linked sites consume internal charge carriers at high temperatures, effectively suppressing charge transfer and thus improving energy storage efficiency.[106] These examples indicate that this patterning method holds promise for bringing innovative technical solutions to the future flexible and printable device industry. Furthermore, this approach could be extended to energy storage devices,[104] 3D printing,[107] and other areas, laying the foundation for the development of various novel device structures.

Incorporating photo-crosslinkers into the dielectric layer plays a pivotal role in enhancing the performance of thin-film capacitors. Key design include the central core and the linking chain segments of the cross-linkers. Current strategies primarily focus on increasing the number of cross-linkable sites in the central core and introducing additional photoactive groups (e.g., each cross-linker containing eight or more photoactive groups). This approach improves cross-linking efficiency, resulting in the formation of denser and more robust films. The increased cross-linking density enhances dielectric properties and boosts energy storage efficiency by reducing the mobility of internal charge carriers.[106]

Based on these works, future research could also explore the design of the linking segments, such as utilizing poly(vinylidene fluoride)[108] or cyano-based polymers, or introducing highly polar groups (e.g., carbon-fluorine bonds or cyano groups) into the linking chain segments. These modifications can enhance the polarization of the film, making the film particularly suitable for electrostatic energy storage applications, where high polarization is essential for maximizing energy efficiency. Furthermore, hydroxyl or acylamide groups could be incorporated into the linking segments of the cross-linkers. These functional groups possess inherent polarity, which is expected to enhance the K-value upon cross-linking, while they can also introduce trapping sites for charge carriers within the dielectric.[109,110] This strategy helps suppress charge carrier motion, further improving the dielectric performance of the films.

7. Other Functions

In addition to the above functions, researchers can enhance other properties of semiconductor devices by selecting or designing appropriate cross-linkers. Functionalized poly(acrylic acid) (pAA) can act as a cross-linker, the cross-linked pAA layer can introduce carboxyl functional groups onto the surface of the organic semiconductor film.[111] (**Figure 10a**) These carboxyl groups then react with the amine groups of phosphatidylethanolamine, enabling the incorporation of phospholipids with specific functions onto the film. This approach creates a platform for recognizing specific components, enabling selective detection of analytes in the low picomolar range. X-ray photoelectron spectroscopy (XPS) confirmed the presence of the PAA coating on the PBTTT surface (Figure 10b). The peak at BE = 289.2 ± 0.2 eV corresponds to the –COOH/R functional group. The figure showed that both cross-linked and non-cross-linked PBTTT/PAA coatings exhibited a substantial amount of –COOH/R functional groups. Additionally, the cross-linked films exhibited higher stability in aqueous solutions. (Figure 10c) After prolonged immersion in water, the –COOH/R functional group content of the non-crosslinked PBTTT/PAA significantly decreased, while the cross-linked PBTTT/PAA showed minimal change. Furthermore, by anchoring a B-PL layer, the detection of streptavidin concentrations as low as 1.6 nM was achieved. (Figure 10d) In addition

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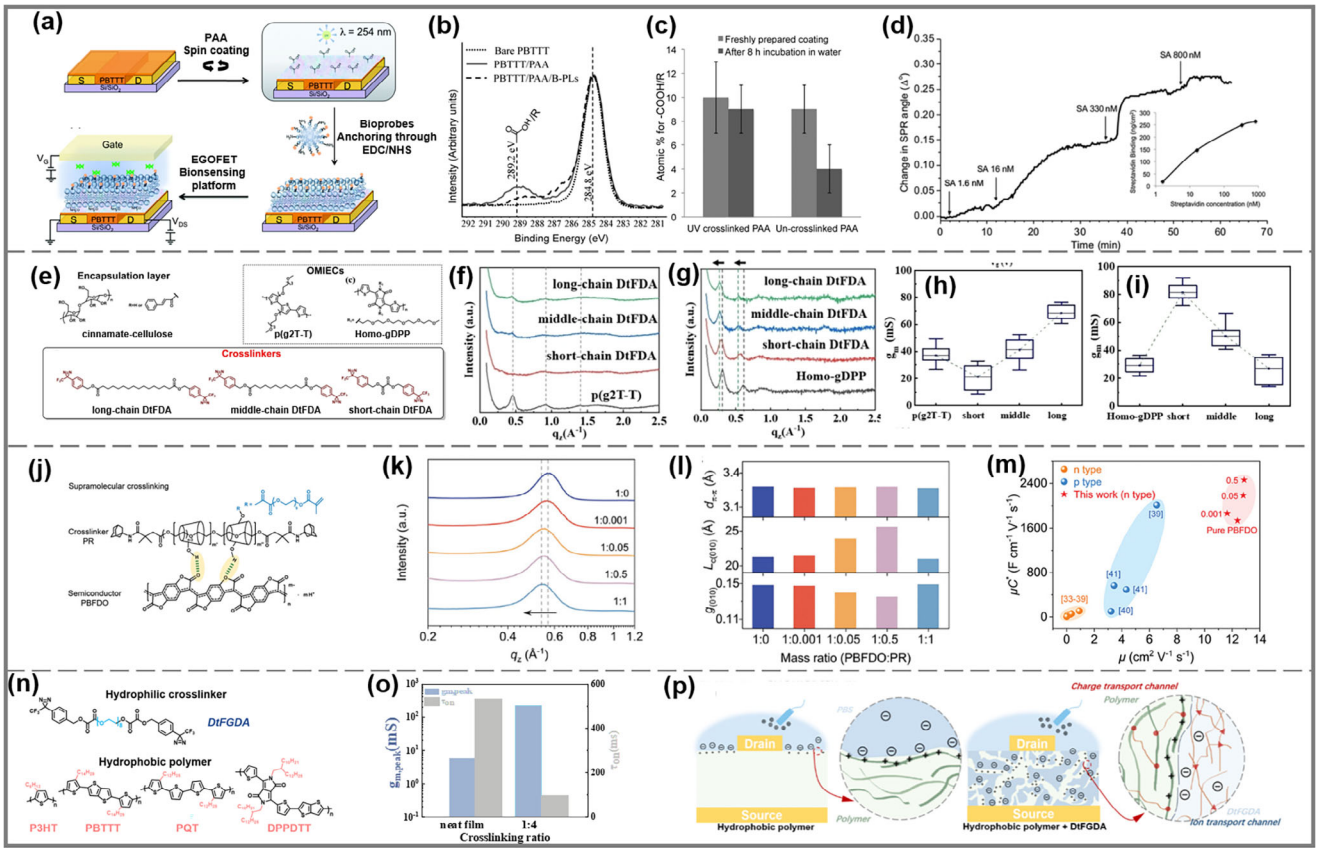


Figure 10. a) Schematic illustration of the EGOFET biosensing platform fabrication process based on PAA. b) C1s XPS region taken from bare PBTTT (dotted line), PBTTT/PAA (solid line) and PBTTT/PAA/B-PLs (dashed line) coatings. c) Relative abundances of –COOH/R functionalities for UV cross-linked and un-crosslinked PAA coatings obtained by XPS measurements immediately after preparation and after incubating the samples for 8 h in water. d) SPR signal response obtained after injecting streptavidin at concentrations ranging from 1.6 to 800 nM on the SPR sensor slide coated with PBTTT/PAA/B-PLs. Reproduced with permission.[111] Copyright 2012, American Chemical Society. Royal Society of Chemistry. e) Chemical structures of p(g2T-T), Homo-gDPP, and DtFDA with three different chain lengths. f) Linecuts for p(g2T-T) : DtFDA blend films (f) and Homo-gDPP : DtFDA blends films g) with different DtFDA intermediate chain lengths. Dependencies of g_m in p-type h) and n-type i) vOECTs on the different chain lengths of DtFDA. Reproduced with permission.[43] Copyright 2023, Wiley-VCH. j) Chemical structures of PBFDO and PR. k) Out-of-plane line cuts show the shift of 100 scattering peaks l) Variation trends in $d_{\pi-\pi}$, $L_{c(010)}$, and $g_{(010)}$ for the pristine PBFDO film and PBFDO:PR films. m) Comparison of μC^* and mobility of PBFDO: PR-based OECTs. Reproduced with permission.[112] Copyright 2025, Wiley-VCH. n) Chemical structures of DtFGDA and polymer. o) Dependencies of gm (blue) and τon (gray) on PBTTT (red) and PBTTT: DtFGDA in vOECTs. p) Schematic working mechanism of neat film and cross-linked film in transistor. Reproduced with permission.[113] Copyright 2025, Wiley-VCH.

to functionalizing the film surface, the cross-linking strategy allows for the modulation of ion-electron coupling. By change the length of the linking segments in the cross-linker, (Figure 10e) a regular impact on the crystal packing of the semiconductors P(g2T-T) and Homo-gDPP was observed (Figure 10f,g).[43] When applied in electrochemical transistors, cross-linked films enable precise patterning while effectively modulating the performance of both P-type and N-type devices. The peak transconductance of P(g2T-T) increased significantly from 21.3 ± 9.8 mS (short-chain DtFDA) to 41.3 ± 9.3 mS (medium-chain DtFDA) and 68.6 ± 5.8 mS (long-chain DtFDA). For Homo-gDPP, the peak transconductance rose from 27.0 ± 10.1 mS (long-chain DtFDA) to 50.3 ± 8.6 mS (medium-chain DtFDA) and 81.6 ± 6.8 mS (short-chain DtFDA) (Figure 10h,i). Using polyrotaxane supramolecular photo-crosslinker facilitates the patterning of poly(benzodifurandione) (PBFDO), (Figure 10j) while serves as a morphology-modulating additive, providing a balanced ion and electron charge transport. After cross-linking, the lattice spacing and coherence length of PBFDO increases (Figure 10k,i). This contributes to enhanced ionic transport of the thin film while maintaining charge mobility[112] (Figure 10m). When ethylene glycol cross-linkers (Figure 10n) are used, the peak transconductance of the PBTTT cross-linked film-based OECT reaches 230.9 mS, which is much higher than that of the neat films. (Figure 10o) The enhanced performance is attributed to a new transport mechanism defined as ionic-electronic separation conduction.[113] (Figure 10p) Additionally, the use of diazirine for cross-linking blue-emitting di-octyl polyfluorene layers significantly reduces film roughness, eliminates defect-induced non-radiative exciton quenching sites, and results in more uniform charge injection and exciton formation.[114]

Consequently, substantial improvements are observed in maximum external quantum efficiency, luminous efficiency, and power efficiency. Moreover, cross-linking generally enhances the

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solvent resistance of films, which facilitates the deposition of multiple organic layers. This improvement significantly optimizes device structures and enables the addition of extra functional layers after cross-linking to further enhance device performance. As photoreactive cross-linkers advance, researchers can achieve patterning while tuning various mechanical and electrical properties of polymers and imparting additional functionalities to films. Anyhow, the films cross-linked with photo-crosslinkers are binary systems, and compared to neat films, the system is more complex to study. Therefore, many studies have focused on developing more efficient cross-linkers that can achieve effective cross-linking with lower cross-linker content and exposure doses.[50,51,115,116] This is highly meaningful, as it holds the potential to further replace commonly organic lithography with direct lithography and enable patterning with higher resolution. However, significant progress is still needed in the development, preparation, and application of cross-linkers, with numerous theoretical and technical challenges awaiting further exploration. The combination of computational chemistry and artificial intelligence-assisted design to identify suitable cross-linker structures for different application scenarios may have significant implications for the development of performance-tailored cross-linkers.

8. Conclusion

Over the past two decades, with the rapid development of organic electronics, direct lithography has matured and made significant progress as a technology highly suitable for patterning organic electronics. The photo-crosslinkers used for direct lithography have evolved from the commercial negative photoresists to a variety of novel molecular structures. Moreover, the resolution of direct lithography has now reached the sub-micrometer level, further enhancing the device density and performance in organic integrated circuits. According to existing reports and our previous work, photo-crosslinkers not only enable precise patterning but also effectively modulate the mechanical and electrical properties of polymer films. This capability has greatly facilitated the widespread application of organic electronics in the fields of flexible, stretchable, and wearable devices, indicating an important future direction for the development of direct lithography technology.

In this review, based on the effects of photo-crosslinkers on organic films, their tensile properties, stability, charge transport, and dielectric properties have been classified and summarized. Concurrently, we look forward to the future design directions of photo-crosslinker and their corresponding development prospects. We hold that variations in the three components of the photo-crosslinkers (the central core, linker chain segments, and photoreactive groups) will have distinct effects on the performance of semiconductors. The central core influences the overall spatial structure of the cross-linker, this could alter the packing of the semiconductor, thus modulating the electronic transport properties of the film. The linker chain segments determine the molecular size of the cross-linker and incorporate various functional groups. By altering the length or changing the functional groups on linker, the intermolecular interactions between cross-linker, or between the cross-linker and the semiconductor, can be modified, which in turn affects the tensile properties of the film. The photoreactive groups in the cross-linker determine the cross-linking process and type, as well as the strength of the cross-linking sites. This influences the stability of the cross-linked network, ultimately affecting the stability and dielectric properties of the film. In this paper, the cross-linkers that can enhance the tensile properties, electronic mobility, stability, and dielectric properties of polymer films are summarized, with an analysis of the underlying reasons for performance enhancement in each case. Furthermore, we have proposed new design concepts for photo-crosslinkers aimed at further enhancing these performance potentials, which will provide stronger support for the development and application of current high-performance organic semiconductor materials. This paper represents the comprehensive summary of introducing cross-linker as a strategy for enhancing the performance of semiconductor films and devices in direct lithography, which could further promote the development and application of direct lithography methods and novel photo-crosslinkers.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

blended film, direct lithography, organic semiconductor, patterning, photo-crosslinkers

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